

Aylin García Soto

 github.com/agarciasoto18 |  [linkedin.com/in/aylin-garcia-soto-02b24a3a](https://www.linkedin.com/in/aylin-garcia-soto-02b24a3a) | 
aylingarciasoto.com |  orcid.org/0000-0001-9828-3229

RESEARCH INTEREST

★ M dwarf magnetic field activity - stellar flares; spectral line emission; photometric modulation; stellar surface feature modeling; rotation periods detection and evolution. ★ Multi-object systems - binaries; exoplanets. ★ Planets and satellites - detection; atmospheres; planet–star interactions

EDUCATION

Dartmouth College	Hanover, NH
Ph.D., Department of Physics and Astronomy	June 2026
M.S., Department of Physics and Astronomy	Mar. 2023
Wesleyan University	Middletown, CT
B.A., Department of Astronomy	May 2018

APPOINTMENTS

Postdoctoral Researcher (w. Prof. Brian Jackson)	State University, <i>present</i>
Graduate Student Research (w. Prof. Elisabeth Newton)	Dartmouth College, 2019-2026
Post-Baccalaureate Research (w. Prof. William Herbst)	Wesleyan University, 2018-2020
McNair Undergrad Research (w. Prof. Edward Moran)	Wesleyan University, 2014-2018
Banneker & Aztlán Summer (w. Prof. Joseph Rodriguez)	Harvard CfA, 2017
KNAC Summer (w. Dr. Steven Souza)	Williams College, 2015

FIRST-AUTHOR PUBLICATIONS

Aylin García Soto, Elisabeth R. Newton, Benjamin V. Rackham, et al. (June 2026). “Revisiting Photometric Variability as a Probe of Spin–Orbit Alignment with Kepler and TESS”. In: *Submitted to The Astrophysical Journal*.

Aylin García Soto, Girish M. Duvvuri, et al. (Apr. 2025). “Short-term Balmer Line Emission Variability in M Dwarfs”. In: 982.2, 98, p. 98. DOI: [10.3847/1538-4357/adb615](https://doi.org/10.3847/1538-4357/adb615). arXiv: [2502.02568](https://arxiv.org/abs/2502.02568) [astro-ph.SR].

Aylin García Soto, Elisabeth R. Newton, Stephanie T. Douglas, et al. (May 2023). “Contemporaneous Observations of H α Luminosities and Photometric Amplitudes for M Dwarfs”. In: 165.5, 192, p. 192. DOI: [10.3847/1538-3881/acc2ba](https://doi.org/10.3847/1538-3881/acc2ba). arXiv: [2304.02543](https://arxiv.org/abs/2304.02543) [astro-ph.SR].

Aylin García Soto, Aleezah Ali, et al. (Apr. 2020). “Evidence for Transparency and Clumps in the Circumbinary Ring of the T Tauri Star V582 Mon (KH 15D)”. In: 159.4, 135, p. 135. DOI: [10.3847/1538-3881/ab6efd](https://doi.org/10.3847/1538-3881/ab6efd).

COLLABORATION PUBLICATIONS

Sophia R. Flury et al. (incl. **Aylin García Soto**) (June 2026). “Activity and Structure of Low Mass Stars: Rossby Number, Mass, and a New Diagnostic”. In: *Resubmitted to The Astrophysical Journal*.

- Yoav Rotman et al. (incl. **Aylin García Soto**) (May 2026). “NASA’s Pandora SmallSat Mission: Simulated Modeling and Retrieval of Near-infrared Exoplanet Transmission Spectra”. In: 171.5, 263, p. 263. DOI: [10.3847/1538-3881/ae4d3c](#). arXiv: [2603.04488](#) [astro-ph.EP].
- Keighley E. Rockcliffe et al. (incl. **Aylin García Soto**) (May 2026). “The unique ability of the Hubble Space Telescope to characterize young exoplanet environments”. In: *arXiv e-prints*, arXiv:2605.28768, arXiv:2605.28768. arXiv: [2605.28768](#) [astro-ph.IM].
- Girish M. Duvvuri et al. (incl. **Aylin García Soto**) (Oct. 2025). “FUMES IV: Optical and Far-ultraviolet Spectra of a Flare on the M Dwarf GJ 4334”. In: 170.4, p. 249. DOI: [10.3847/1538-3881/adff74](#). arXiv: [2508.18459](#) [astro-ph.SR].
- Emily M. Boudreaux et al. (incl. **Aylin García Soto**) (Apr. 2024). “Correlations between Ca II H and K Emission and the Gaia M Dwarf Gap”. In: 965.1, p. 56. DOI: [10.3847/1538-4357/ad2e8c](#). arXiv: [2402.14984](#) [astro-ph.SR].
- Michael Fausnaugh et al. (incl. **Aylin García Soto**) (Sept. 2021). “The TESS Mission Target Selection Procedure”. In: 133.1027, p. 095002. DOI: [10.1088/1538-3873/ac1d3f](#). arXiv: [2109.02665](#) [astro-ph.IM].
- Natalia M. Guerrero et al. (incl. **Aylin García Soto**) (June 2021). “The TESS Objects of Interest Catalog from the TESS Prime Mission”. In: 254.2, p. 39. DOI: [10.3847/1538-4365/abefe1](#). arXiv: [2103.12538](#) [astro-ph.EP].
- Pamela Rowden et al. (incl. **Aylin García Soto**) (Aug. 2020). “TIC 278956474: Two Close Binaries in One Young Quadruple System Identified by TESS”. In: 160.2, p. 76. DOI: [10.3847/1538-3881/ab9d20](#). arXiv: [2006.08979](#) [astro-ph.SR].
- H. Parviainen et al. (incl. **Aylin García Soto**) (Jan. 2020). “MuSCAT2 multicolour validation of TESS candidates: an ultra-short-period substellar object around an M dwarf”. In: 633, A28. DOI: [10.1051/0004-6361/201935958](#). arXiv: [1911.04366](#) [astro-ph.EP].
- Daniel Huber et al. (incl. **Aylin García Soto**) (June 2019). “A Hot Saturn Orbiting an Oscillating Late Subgiant Discovered by TESS”. In: 157.6, p. 245. DOI: [10.3847/1538-3881/ab1488](#). arXiv: [1901.01643](#) [astro-ph.EP].
- Joseph E. Rodriguez et al. (incl. **Aylin García Soto**) (May 2019). “An Eccentric Massive Jupiter Orbiting a Subgiant on a 9.5-day Period Discovered in the Transiting Exoplanet Survey Satellite Full Frame Images”. In: 157.5, p. 191. DOI: [10.3847/1538-3881/ab11d9](#). arXiv: [1901.09950](#) [astro-ph.EP].
- Diana Dragomir et al. (incl. **Aylin García Soto**) (Apr. 2019). “TESS Delivers Its First Earth-sized Planet and a Warm Sub-Neptune”. In: 875.2, p. L7. DOI: [10.3847/2041-8213/ab12ed](#). arXiv: [1901.00051](#) [astro-ph.EP].

HONORS & AWARDS

★ Dartmouth College Conference Graduate Student Travel Award	2025
★ Matt Johns Astronomy Graduate Student Fund	2024
★ The Selamawit Tsehaye Teaching Award	2021
★ NSF GRFP Honorable Mention	2021
★ E.E. Just Liftoff Fellow	2019
★ AAS Famous Grant	2019
★ Wesleyan University Honors Thesis	2018
★ Wesleyan University Department of Astronomy's Littell Prize	2018
★ Ronald E. McNair Scholars Program	2015-2018
★ NASA CT Space Grant Scholarship	2015
★ Janey Scholars Program	2014-2018

TELESCOPE PROPOSALS & OBSERVING EXPERIENCE

JWST Observing Proposal: (PI: Benjamin Rackham; Co-I: Aylin García Soto)	Oct. 2025
NSF NOIRLab Observing Proposal: (PI: Sarah Logsdon; Co-I: Aylin García Soto)	Sept. 2025
TESS Guest Investigator Proposal: <i>Filling In The Spots: Multi-color Analysis Of M Dwarf Photospheres</i> (PI: William Waalkes; Co-I: Aylin García Soto)	Nov. 2024
Observing Time: MDM McGraw-Hill 1.3m Telescope (PI: Aylin García Soto)	2019-2024 29 Nights
MDM Hiltner 2.4m Telescope (PI: Aylin García Soto)	32 Nights

RESEARCH TALKS

★ Seminars & Colloquia

- **University of Arizona Origins Seminar** ([Link](#)), “Magnetic Fields Activity on M-type Cool Stars Using TESS.” October 6, 2025. Host: Prof. Brittany Miles.
- **Black Space Week: AAS Journal Author Series** ([Link](#)), “AAS Journal Author Series: Aylin Garcia Soto on 2025ApJ...982...98G.” June 12, 2025. Host: Frank Timmes.
- **Boston University Astronomy Colloquium**, “Simultaneous Observations with TESS and OSMOS: Magnetic Fields Activity on M-type Cool Stars.” March 17, 2025. Host: Prof. Catherine Espilliat.
- **University of Michigan APS Seminar**, “Fully Convective M Dwarfs: Variability in Magnetically Sensitive Balmer Emission Lines.” García Soto, A., October 1, 2024. Host: Prof. Lia Corrales.
- **University of Florida Astronomy Colloquium**, “Simultaneous Observations with TESS and OSMOS: Magnetic Fields Activity on M-type Cool Stars.” August 29, 2024. Host: Prof. Sarah Ballard.
- **University of Colorado Boulder APS Seminar**, “Contemporaneous Observations of H α Luminosities and Photometric Amplitudes for M dwarfs.” June 23, 2023. Host: Prof. Zachory Berta-Thompson.
- **TESS Mission Update Meeting (Science Team Meeting)**, “Contemporaneous Observa-

tions of $H\alpha$ Luminosities and Photometric Amplitudes for M Dwarfs.” 2023. MIT Building 6, Room 120.

★ Contributed Talks

- **MIT Stars & Planets Workshop**, “TESS Selection Affects and Stellar Obliquities in Fully Convective M dwarf Transit Hosts.”
- **MIT Stellar Contamination Workshop**, “Fully Convective M Dwarfs: Variability in Magnetically Sensitive Balmer Emission Lines.”
- **TESS Science Conference 3 (TSC3)**, “Fully Convective M Dwarfs: Variability in Magnetically Sensitive Balmer Emission Lines.” <https://zenodo.org/records/13136772>

★ Posters

- **Know Thy Star, Know Thy Planet 2 Conference**, “M Dwarfs Variability in Magnetically Sensitive Balmer Emission Lines.” García Soto, A. et al. 2025.
- **Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun. 22.**, “M Dwarfs Variability in Magnetically Sensitive Balmer Emission Lines.” García Soto, A. et al. 2024. <https://zenodo.org/records/13118976>
- **Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun. 21.**, “Contemporaneous Observations of $H\alpha$ Luminosities and Photometric Amplitudes for M dwarfs.” García Soto, A. et al. 2022. <https://zenodo.org/record/7523687>
- **233rd Meeting of the American Astronomical Society**, “Different Observational Properties of the Leading and Trailing Edges of the KH 15D Circumbinary Ring”, García Soto, A. et al. 2019.
- **231st Meeting of the American Astronomical Society**, “Photometric Follow-up of Eclipsing Binary Candidates from KELT and Kepler”, García Soto, A. et al. 2018.
- **231st Meeting of the American Astronomical Society**, “Uncovering extreme AGN variability in serendipitous X-ray source surveys”, E. C. Moran, García Soto, A. et al. 2018.
- **APS Conference for Undergraduate Women in Physics at Harvard**, “Dramatic X-ray Variability in AGNs: New Insights into Black Hole Fueling.” García Soto, A., and Moran, E. C. 2017.

NASA MISSION MEMBERSHIP

NASA SmallSat Pandora Science Team Member	Sept. 2025 - Present
NASA/MIT TESS SG1 Team Member	June 2026 - Present
NASA/MIT TESS Science Team Member	Sept. 2019 - Present

ADVISING & MENTORING

The Short Period Planets Group Mentor <i>Mentoring Boise State University Undergrads in Research</i>	Mar. 2026 - Present
Graduate Student Mentor for Dartmouth IGMP <i>Program for Mentoring International Graduate Students.</i>	Sept. 2025 - June 2026
Graduate Student Mentor for GSEC <i>Program for mentoring first-year latine Graduate Students.</i>	Sept. 2024 - June 2026
Penpal for Letters for Pre-Scientist (LPS) Penpals with middle schoolers in California.	Sept. 2023 - May 2026
Chase Alvarado-Anderson <i>Undergraduate Student (Dartmouth), now Graduate Student (Standford)</i>	2020 - June 2026
Dartmouth ASURE Graduate Student Panel Panelist for discussion of graduate school process	July 2022
Vera Krumm <i>Undergraduate Student at Dartmouth College</i>	(2021-2022)
Abigail Burrows <i>Undergraduate Student at Dartmouth College</i>	(2020-2023)
Graduate Student Mentor for E.E. Just Program <i>Liftoff Fellow - informal undergraduate mentor</i>	Sept. 2020 - Present
Women+ of Color Project Panels <i>The Path to Graduate School and Non-Traditional Routes & Internships</i>	2019-2020
Graduate Student Mentor for the McNair Program <i>McNair Fellow - informal undergraduate mentor</i>	Sept. 2016 - Sept. 2018

TEACHING APPOINTMENTS

Astro 61: Observational Techniques in Astronomy (Week 7/Day 2)	Guest Lecturer, <i>Fall 2025</i>
Astro 61: Observational Techniques in Astronomy	Teaching Assistant, <i>Fall 2025</i>
Astro 1 Exploration of the Solar System (Week 8/Day 3) <i>Intro to Astronomy Part 1, Nonmajors</i>	Guest Lecturer, <i>Summer 2025</i>
Astro 1 Exploration of the Solar System (Week 8/Day 1) <i>Intro to Astronomy Part 1, Nonmajors</i>	Guest Lecturer, <i>Summer 2025</i>
Astro 15: Stars and the Milky Way <i>Intro to Astronomy, Majors</i>	Teaching Assistant, <i>Spring 2025</i>
Astro 61: Observational Techniques in Astronomy	Teaching Assistant, <i>Fall 2023</i>
Astro 74: Astrophysics <i>Radiative Processes</i>	Teaching Assistant, <i>Fall 2022</i>
Astro 2: Exploring the Universe <i>Intro to Astronomy Part 2, Nonmajors</i>	Teaching Assistant, <i>Summer 2021</i>
Astro 15: Stars and the Milky Way <i>Intro to Astronomy, Majors</i>	Teaching Assistant, <i>Spring 2021</i>
Astro 2/3: Exploring the Universe, with Laboratory <i>Intro to Astronomy Part 2, Nonmajors</i>	Teaching Assistant, <i>Fall 2020</i>
Astro 2: Exploring the Universe <i>Intro to Astronomy Part 2, Nonmajors</i>	Grader, <i>Summer 2020</i>
Astro 15: Stars and the Milky Way <i>Intro to Astronomy, Majors</i>	Teaching Assistant, <i>Spring 2020</i>
Phys 3: General Physics I <i>Intro to Physics for Astronomy Majors</i>	Teaching Assistant, <i>Winter 2020</i>

SERVICE & OUTREACH EXPERIENCE

Unistellar Citizen Science Flares Project June 2026 - Present

- Managed Idaho Teachers with taking observations using Unistellar telescopes.
- Loaned Unistellar Telescopes to Idaho Teachers for Research and Lesson Plans

Black In Space Week 2026 June 2026

- Evaluated Lighting Talks for the Graduate and Undergraduate in a Research Showcase
- Wrote an Astrobites entry on Little Red Dots: “Little Red (Dot), what a red, old, compact galaxy you are. All the better to confuse you with!”

IDAstro Outreach Mentor Mar. 2026 - Present

- Volunteered at the Bruneau Dunes State Park Observatory and presented research.
- Mentored undergraduates from near Idaho Colleges conducting observations

Dartmouth Outreach Event Organizer Sept. 2019 - Mar. 2026

- *Planned educational activities for Science Day and the Astronomy Kick-Off Event.*
- *Co-lead the Montshire Museum of Science Astronomy Science Day Events.*

Dartmouth Telescope Operator and Trainer Sept. 2019 - Mar. 2026

- *Operated the 14-inch Meade telescope for public observation nights.*
- *Trained astronomers and students on the MDM Hiltner 2.4m and McGraw-Hill 1.3m telescopes in Arizona - 13 Nights*

Wikipedia Page KH 15D (V 582 Mon) June 2018

- *Created Wikipedia page for the T Tauri Binary System KH 15D ([Link](#))*

Wesleyan Outreach Organizer for Kids' Night and Space Night Sept. 2015 - Sept. 2017

- *Organized Astronomical presentations – 16 Kids' Nights and 4 Space Nights*
- *Operated of the 16-inch Meade Telescope and the 24-inch Perkin Telescope.*
- *Created hands-on activities for the Middletown community and their families.*

Wesleyan Science Outreach Co-Coordinator Sept. 2015 - Sept. 2017

- *Co-instructed the pedagogical course CHEM242.*
- *Recruited of volunteers*
- *Created and Taught club leaders how to create lesson plans for school visits and Science Saturday.*

PROFESSIONAL EXPERIENCE

NASA ATP Executive Secretary	Virtual, 2023
Graduate Student Council Representative/Leader	Dartmouth College, 2020-2023
TESS Objects of Interest Deputy Manager	MIT, 2018-2019
Prof. Sara Seager's Personal Assistant	MIT, 2018-2019

SOFTWARE CONTRIBUTION

- ★ `protify` - TESS rotation periods
- ★ `starrotate` - TESS rotation periods
- ★ `TESS-ExoClass` - TESS Robo vetter

REFERENCES

Brian Jackson (email: bjackson@boisestate.edu)

Professor of Physics and Astronomy

Boise State University

Elisabeth R. Newton (email: Elisabeth.R.Newton@dartmouth.edu)

Professor of Physics and Astronomy

Dartmouth College

Stephanie T. Douglas (email: StephanieTDouglas@gmail.com)

Assistant Professor of Physics

Lafayette College

Zachory Berta-Thompson (email: zach.bertathompson@colorado.edu)

Professor of Astrophysics and Planetary Science

University of Colorado, Boulder